

LECTURE 11 · DISCRETE MATHEMATICS

Sets turn rules into collections

What belongs? What overlaps? What changes?

Three requests. How many permissions?

An app requests camera, location, camera.

$\{\text{camera, location, camera}\}$

$\{\text{camera, location}\}$: **two permissions.**

A repeated element adds nothing.

A set records membership

An unordered collection of distinct objects.

Belongs

$$a \in A$$

Does not belong

$$a \notin A$$

$$\{1, 2, 2, 3\} = \{3, 1, 2\}$$

Empty is different from containing empty

No elements

$$\emptyset = \{\}, \quad |\emptyset| = 0$$

One element

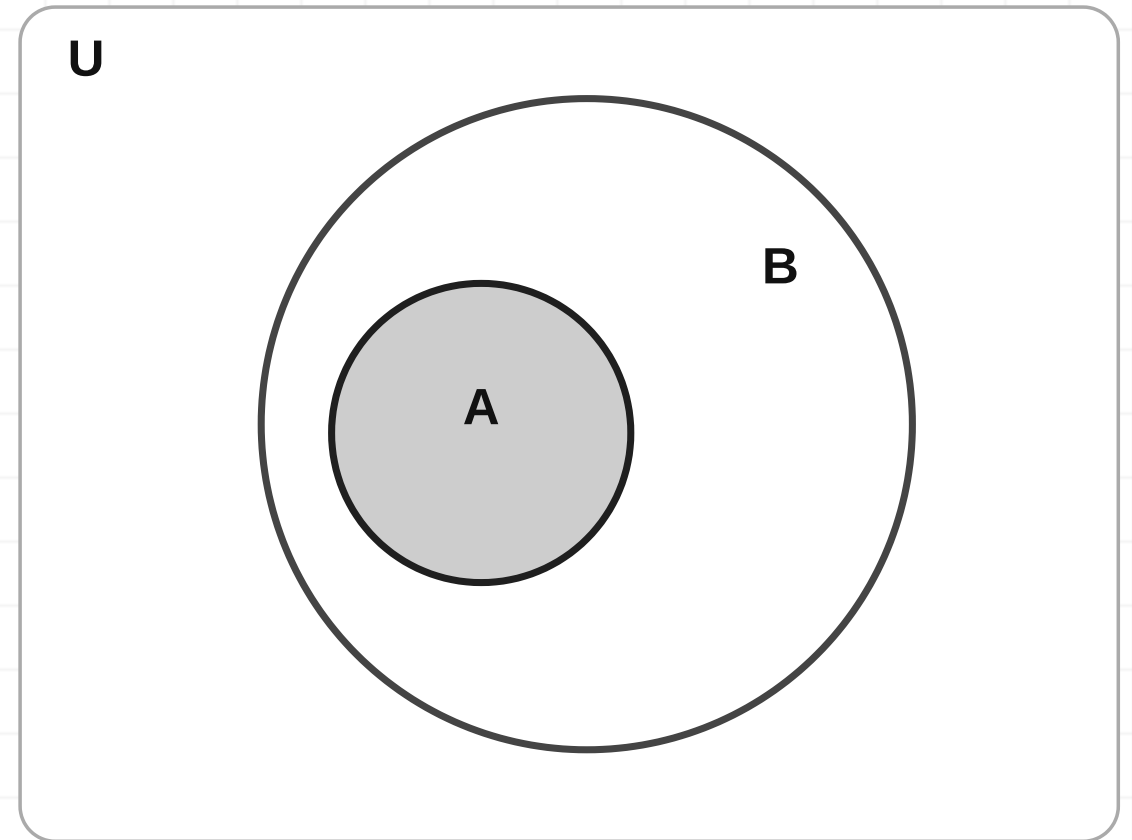
$$\{\emptyset\}, \quad |\{\emptyset\}| = 1$$

A singleton has exactly one element.

A subset keeps every element

Every set has $\emptyset$ and itself as subsets. A proper subset also leaves something out.

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B)$$



$$A \subsetneq B \iff A \subseteq B \text{ and } A \neq B$$

Is it listed? Or is every piece listed?

Let $T = \{0, \{0\}, \{1\}\}$. True or false?

01 $\{0\} \in T$

True: $\{0\}$ is one of the three listed elements.

02 $\{0\} \subseteq T$

True: its only element, 0, is listed.

03 $\{1\} \subseteq T$

False: $1 \notin T$. Only $\{1\} \in T$.

04 $\emptyset \in T$ and $\emptyset \subseteq T$

False, then true. $\emptyset$ is a subset of every set, but an element only when listed.

$X \in T$: is X itself listed? $X \subseteq T$: is each element of X listed?

The power set collects every subset

$\mathcal{P}(S) = \{A : A \subseteq S\}$. Let $S = \{0, 1\}$.

$$\mathcal{P}(S) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$$

$$|S| = n \Rightarrow |\mathcal{P}(S)| = 2^n. \quad \emptyset \in \mathcal{P}(S), \quad S \in \mathcal{P}(S).$$

A power set can contain sets of sets

Let $R = \{1, \{1\}\}$.

01 **Find** $|R|$ **and** $|\mathcal{P}(R)|$.

$|R| = 2$, so $|\mathcal{P}(R)| = 2^2 = 4$.

02 **List** $\mathcal{P}(R)$.

$\{\emptyset, \{1\}, \{\{1\}\}, \{1, \{1\}\}\}$

03 **Is** $\{1\} \in \mathcal{P}(R)$? **Is** $\{\{1\}\} \in \mathcal{P}(R)$?

Both: $\{1\} \subseteq R$ **and** $\{\{1\}\} \subseteq R$.

$$X \in \mathcal{P}(R) \iff X \subseteq R$$

Five operations. Five membership rules.

$$U = \{1, \dots, 6\}, \quad A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5\}.$$

Operation	x belongs when	Example
Union $A \cup B$	$x \in A \vee x \in B$	$\{1, 2, 3, 4, 5\}$
Intersection $A \cap B$	$x \in A \wedge x \in B$	$\{3, 4\}$
Difference $A \setminus B$	$x \in A \wedge x \notin B$	$\{1, 2\}$
Symmetric difference $A \triangle B$	x is in exactly one	$\{1, 2, 5\}$
Complement A^c	$x \in U \wedge x \notin A$	$\{5, 6\}$

Rosen writes $A \oplus B$ for $A \triangle B$ and $A - B$ for $A \setminus B$.

EXERCISE 3 · TRY FIRST

Compute each set. Then check with bits.

$$U = \{1, \dots, 8\}, \quad A = \{1, 2, 3, 6\}, \quad B = \{2, 4, 6, 8\}.$$

01 $A \cup B, \quad A \cap B, \quad B \setminus A, \quad (A \cup B)^c$
 $\{1, 2, 3, 4, 6, 8\}, \quad \{2, 6\}, \quad \{4, 8\}, \quad \{5, 7\}$

02 **Bit strings for A and B , in the order $1, \dots, 8$.**
 $A = 11100100, \quad B = 01010101$

03 **Bitwise AND and XOR. Decode both.**
 $01000100 = \{2, 6\} = A \cap B; \quad 10110001 = \{1, 3, 4, 8\} = A \triangle B$

Bit i is 1 exactly when element i belongs. OR, AND, XOR = union, intersection, symmetric difference.

Prove set equality from both sides

Claim: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Translate to logic

$$x \in A \wedge \neg(x \in B \vee x \in C)$$

De Morgan: $x \in A, x \notin B, x \notin C$.

Left $\subseteq$ right

$$x \in A \setminus B \wedge x \in A \setminus C$$

Take any x on the left; the three facts put it in both differences.

Right $\subseteq$ left

$$x \in A, x \notin B, x \notin C$$

So $x \notin B \cup C$, and $x \in A \setminus (B \cup C)$.

$$X = Y \iff X \subseteq Y \text{ and } Y \subseteq X$$

A membership table checks all eight cases

1 = x is in the set. 0 = x is not.

A	B	C	$B \cup C$	$A \setminus (B \cup C)$	$A \setminus B$	$A \setminus C$	$(A \setminus B) \cap (A \setminus C)$
1	1	1	1	0	0	0	0
1	1	0	1	0	0	1	0
1	0	1	1	0	1	0	0
1	0	0	0	1	1	1	1
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Columns 5 and 8 agree in every row, so the two sets are equal.

Prove it for every set, or break it once

Decide. Then give a proof or one checked counterexample.

01 If $A \setminus C = B \setminus C$, must $A = B$?

No. $A = \{1\}$, $B = \emptyset$, $C = \{1\}$: both differences are $\emptyset$.

02 If $A \subseteq B$, must $\mathcal{P}(A) \subseteq \mathcal{P}(B)$?

Yes. $X \in \mathcal{P}(A) \Rightarrow X \subseteq A \subseteq B \Rightarrow X \in \mathcal{P}(B)$.

03 If $A \triangle C = B \triangle C$, must $A = B$?

Yes. $\triangle C$ undoes itself: $A = (A \triangle C) \triangle C = (B \triangle C) \triangle C = B$.

A false claim needs one specific example. A true claim needs an arbitrary element.

With three sets, restore the triple overlap

Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$. Three sets need one more correction.

Add the sets

$$|A| + |B| + |C|$$

Subtract pairwise overlaps

$$-|A \cap B| - |A \cap C| - |B \cap C|$$

Add the triple overlap

$$+|A \cap B \cap C|$$

The result is $|A \cup B \cup C|$. A triple member counts $3 - 3 + 1 = 1$ time.

Count multiples with inclusion–exclusion

In $\{1, \dots, 100\}$, let D_m be the set of multiples of m .

$$|D_m| = \lfloor 100/m \rfloor, \quad D_2 \cap D_3 = D_6$$

01 **How many are divisible by 2, 3, or 5?**

$$|D_2 \cup D_3 \cup D_5| = 50 + 33 + 20 - 16 - 10 - 6 + 3 = 74.$$

02 **How many are divisible by exactly one of them?**

Only 2: $50 - 16 - 10 + 3 = 27$. **Only 3:** 14. **Only 5:** 7. **Total** 48.

Work outward from the centre: triple overlap, then pairs, then single sets.

List the elements. Or give the rule.

Roster

$$E = \{0, 2, 4, 6\}$$

Set builder

$$E = \{2k : k \in \mathbb{Z}, 0 \leq k \leq 3\}$$

The colon means “such that”.

The familiar number sets

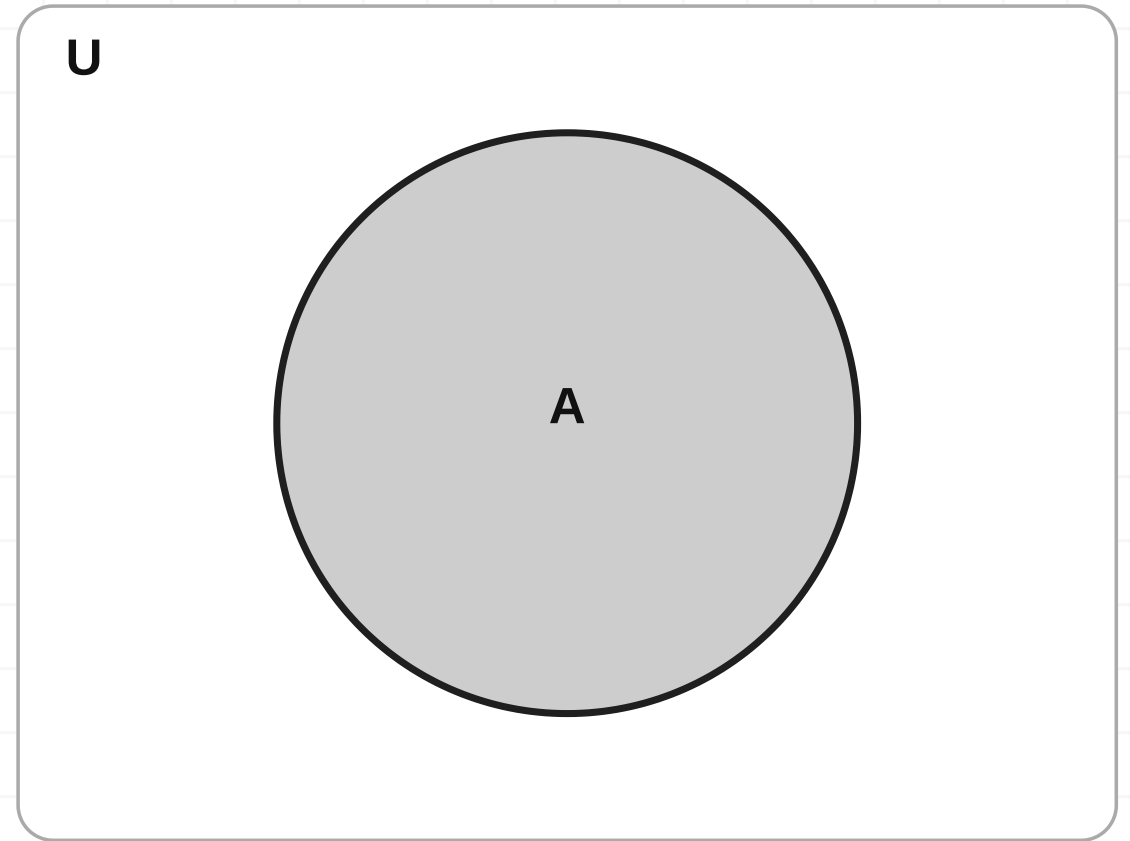
Set	Name	Members
$\mathbb{N}$	Natural numbers	$0, 1, 2, \dots$
$\mathbb{Z}$	Integers	$\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Z}^+$	Positive integers	$1, 2, 3, \dots$
$\mathbb{Q}$	Rational numbers	$p/q : p, q \in \mathbb{Z}, q \neq 0$
$\mathbb{R} ; \mathbb{R}^+$	Reals; positive reals	$\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$
$\mathbb{C}$	Complex numbers	$a + bi : a, b \in \mathbb{R}, i^2 = -1$

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$$

Choose the universe first

U contains every object currently under discussion.

$$A \subseteq U$$



Vowels: letters. App permissions: available permissions.

One rewrite proves the inclusion

$$A = \{6r + 12 : r \in \mathbb{Z}\}, \quad B = \{3s : s \in \mathbb{Z}\}.$$

$$A \subseteq B \quad \text{but} \quad B \not\subseteq A$$

$$x = 6r + 12 = 3(2r + 4) \Rightarrow x \in B.$$

$3 \in B$, but $3 \notin A$: **every element of A is divisible by 6.**

An arbitrary element proves inclusion. One counterexample refutes it.

Different rules. The same set?

$$A = \{2r : r \in \mathbb{Z}\}, \quad B = \{2s - 2 : s \in \mathbb{Z}\}.$$

$$A = B \iff A \subseteq B \wedge B \subseteq A$$

$$2r = 2(r + 1) - 2 \Rightarrow A \subseteq B.$$

$$2s - 2 = 2(s - 1) \Rightarrow B \subseteq A.$$

Each new element doubles the choices

Add $x \notin S$. Each old subset A gives two choices.

$$|S| = n \implies |\mathcal{P}(S)| = 2^n$$

Leave it out

A

Put it in

$A \cup \{x\}$

Include or exclude each element.

How many permission bundles?

Available permissions: camera, location, microphone. Any subset is allowed.

01 **How many bundles?**

$$2^3 = 8.$$

02 **How many contain camera?**

$$2^2 = 4: \text{choose the other two freely.}$$

Two translucent filters overlap above a collection of objects. Connect the operations to familiar logical connectives.

ACT II · SET OPERATIONS

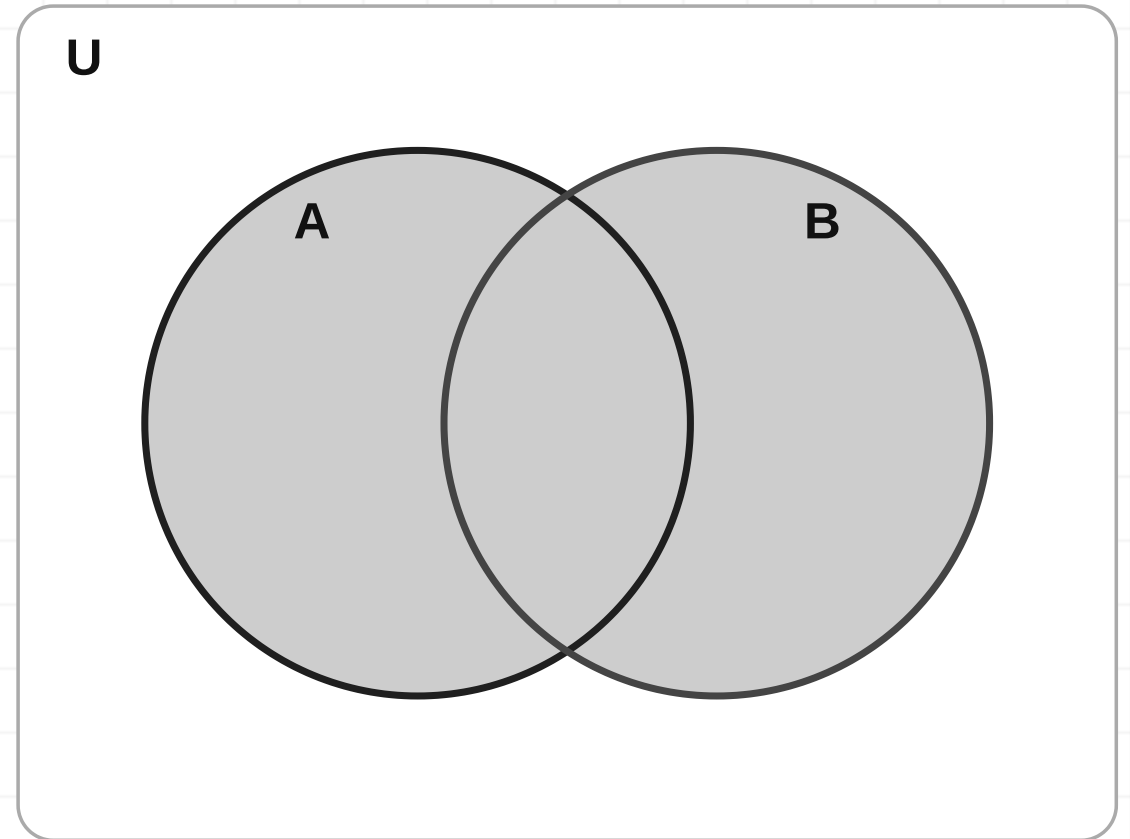
Combine membership rules

OR. AND. NOT. Now applied to collections.

Union means at least one

Photos tagged “night” OR “city”.

$$x \in A \cup B \iff x \in A \vee x \in B$$

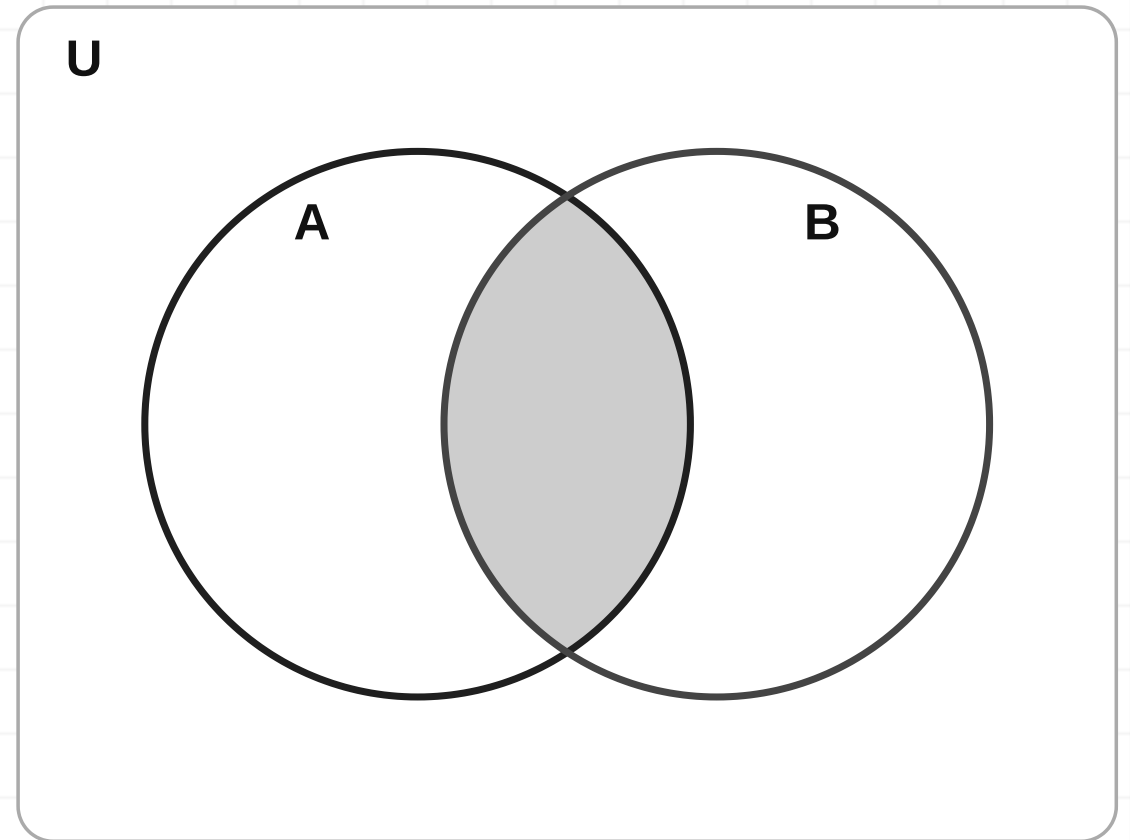


The overlap is included.

Intersection means both

Photos tagged “night” AND “city”.

$$x \in A \cap B \iff x \in A \wedge x \in B$$

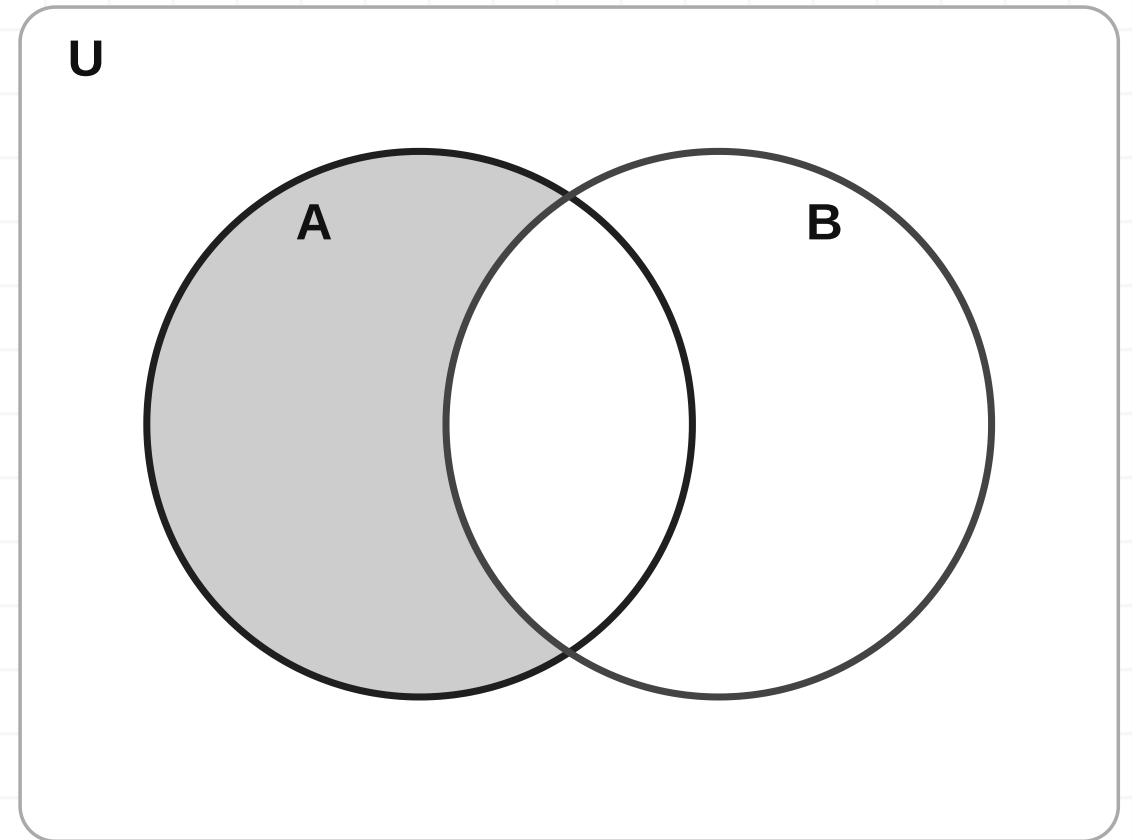


Only the overlap survives.

Difference removes the second set

$$A = \{1, 3, 5\}, B = \{1, 2, 3\}$$

$$A \setminus B = \{x \in A : x \notin B\}$$

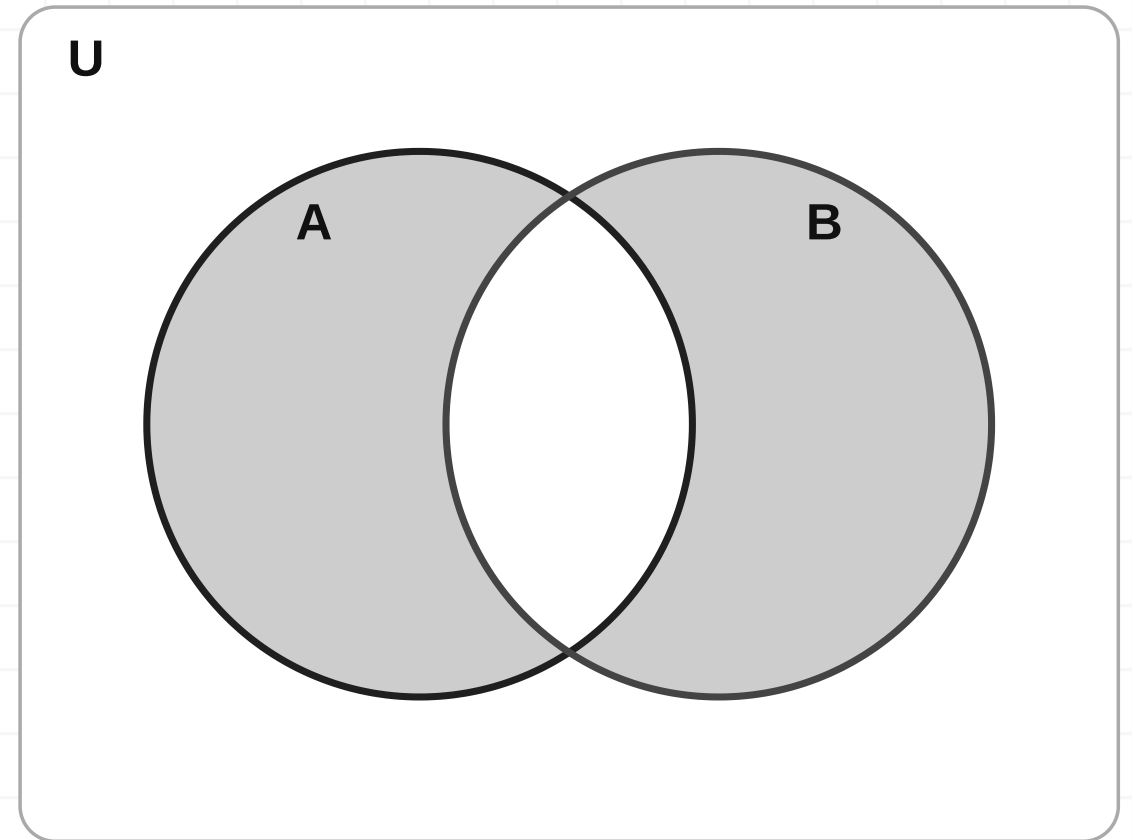


$$A \setminus B = \{5\}, \quad B \setminus A = \{2\}$$

Symmetric difference means exactly one

One tag, but not both.

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$



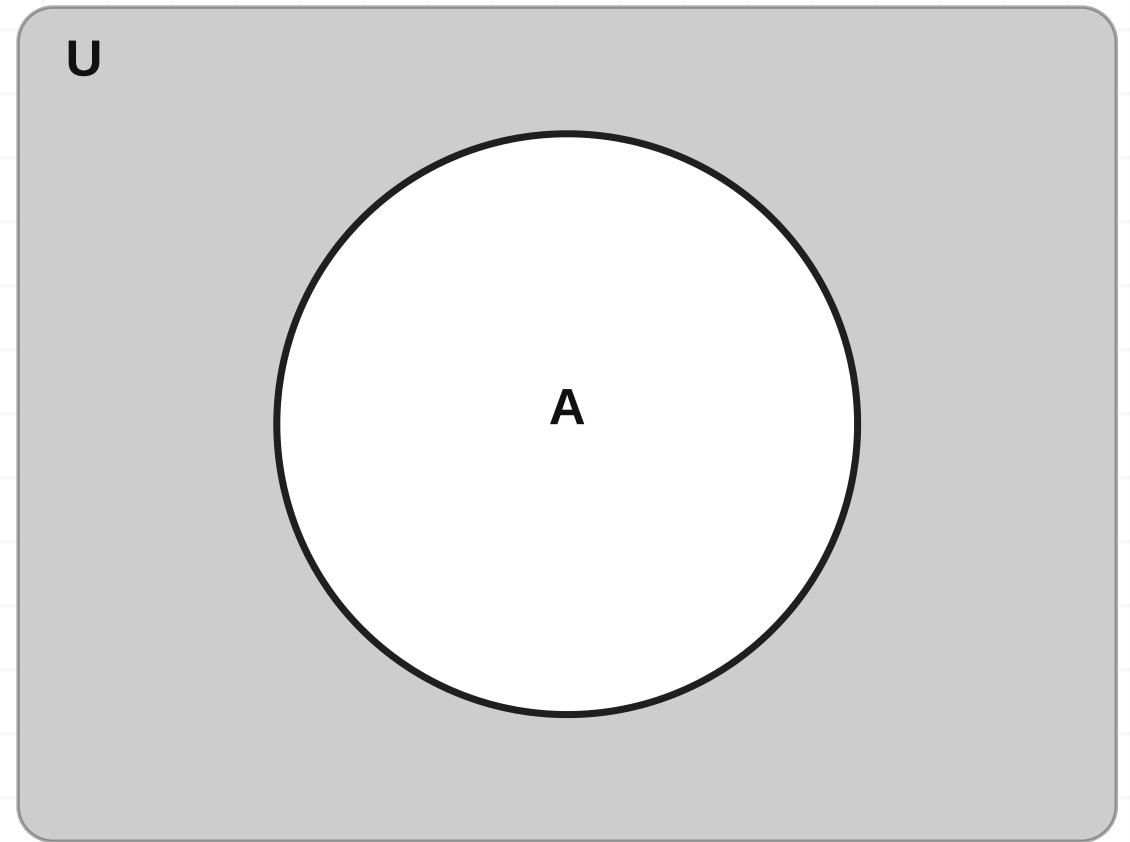
$$A \triangle A = \emptyset,$$

$$A \triangle A \triangle A = A$$

Complement depends on the universe

$$U = \{1, 2, 3, 4\}, \quad A = \{1, 3\}.$$

$$A^c = U \setminus A$$

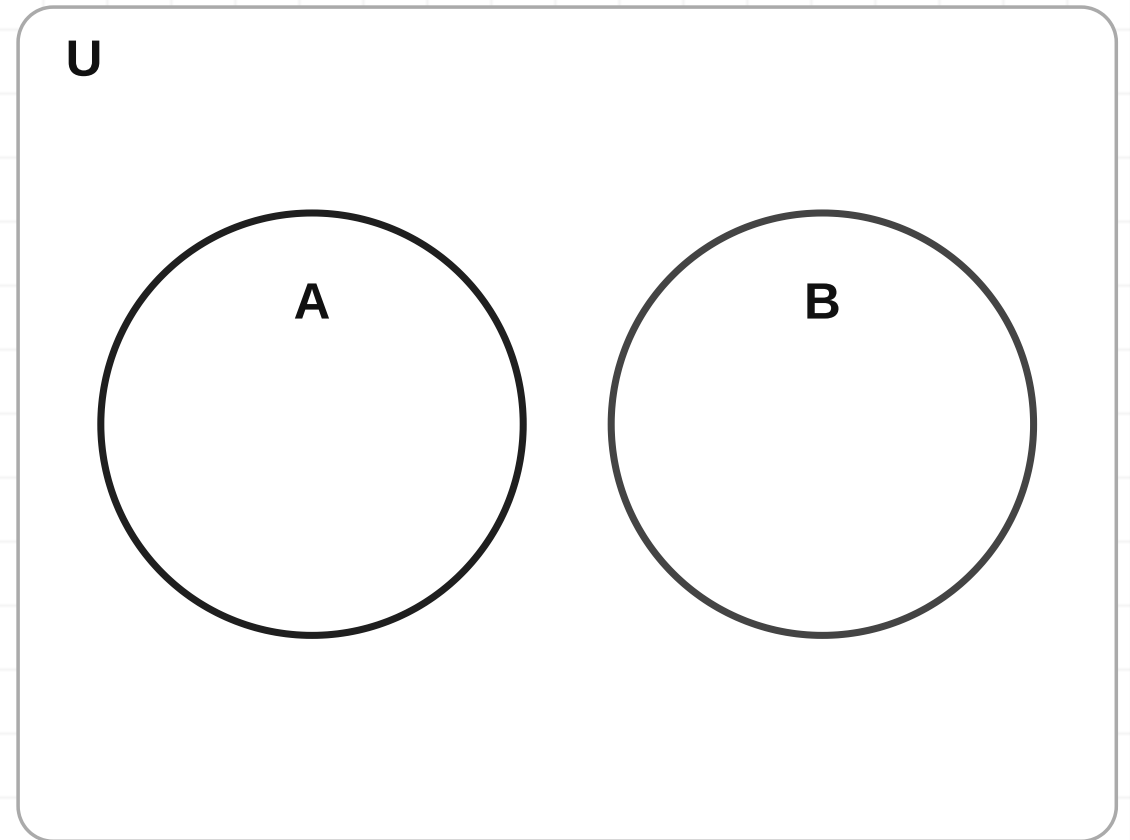


$A^c = \{2, 4\}$. Change U ; the answer may change.

Disjoint means no shared elements

Even IDs and odd IDs never overlap.

$$A \cap B = \emptyset$$



$\emptyset$ is disjoint from every set, including itself.

Prove a set law one element at a time

Fix an arbitrary element x in U .

Complement of intersection

$$x \in (A \cap B)^c \iff \neg(x \in A \wedge x \in B)$$

Use De Morgan

$$\iff x \notin A \vee x \notin B$$

Translate back

$$\iff x \in A^c \cup B^c$$

$$(A \cap B)^c = A^c \cup B^c$$